

Mathematics Stage 4

Introduction to Algebra

Book 1

Version: 260930

Feedback:
<https://MrDingMaths.com>

Contents

Syllabus Content.....	2
Generalising Relationships using Algebra.....	3
Algebraic Conventions	6
Parts of an Expression	11
Forming Expressions from Diagrams	16
Forming Expressions from Descriptions.....	21
Substitution and Evaluation.....	28
Substitution with Negative Numbers	37
Generating Patterns by Substitution.....	43

Syllabus Content

MA4-ALG-C-01 generalises number properties to operate with algebraic expressions including expansion and factorisation

Examine the concept of pronumerals as a way of representing numbers

- Examine and recognise that pronumerals can be used to represent one or more numerical values and when pronumerals have more than one numerical value, they may then be referred to as variables
- Identify and define an algebraic expression as an expression formed by combining numbers and algebraic symbols using arithmetic operations
- Use concise algebraic notation and conventions for multiplication, division and powers, and explain the meanings for each convention

Create algebraic expressions and evaluate them by substitution

- Generate algebraic expressions by translating descriptions and vice versa
- Substitute numbers into algebraic expressions and evaluate the result
- Generate a number pattern from an algebraic expression

Generalising Relationships using Algebra

Algebra

- **Algebra** communicates patterns, rules, and relationships involving numbers.
- **Pronumerals** are letters or symbols that represent numbers, such as x , y or π .
- **Variables** are pronumerals with many possible numerical values.

Generalising

- An **algebraic expression** combines numbers, pronumerals, and symbols.
- Algebra shows relationships that work for any number, not just one specific example.
- Showing a pattern works for any number is called **generalising**.

Example Generalise rules of arithmetic

Look at the pattern and generalise a rule.

$$\begin{aligned}5 \times 0 &= 0 \\6 \times 0 &= 0 \\51 \times 0 &= 0 \\\frac{1}{2} \times 0 &= 0\end{aligned}$$

English expression: **Any number times zero is zero**

Guided Practice

Generalise these relationships.

a

$$\begin{aligned}4 \times 1 &= 4 \\7 \times 1 &= 7 \\(-3) \times 1 &= -3\end{aligned}$$

English expression:

.....

Any number multiplied by 1 is itself.

Algebraic expression:

$n \times 1 = n$

b

$$\begin{aligned}9 + 9 + 9 &= 3 \times 9 \\2 + 2 + 2 &= 3 \times 2 \\(-4) + (-4) + (-4) &= 3 \times (-4)\end{aligned}$$

English expression:

.....

Any number added to itself twice is three times the number.

Algebraic expression:

$n + n + n = 3 \times n$

Key Ideas

1. Pronumerals are letters or symbols that represent
2. If a pronumeral represents *any* number, then we call it a
3. Algebra involves using to represent numbers so we can quickly communicate mathematical relationships.

1 Generalise these number facts.

a

$$6 - 6 = 0$$

$$1 - 1 = 0$$

$$17 - 17 = 0$$

English

expression: Any number subtract by itself is zero.

Algebraic

expression:
 $n - n = 0$

b

$$9 + 9 = 2 \times 9$$

$$3 + 3 = 2 \times 3$$

$$(-4) + (-4) = 2 \times (-4)$$

English

expression:

 Any number added to itself is two times the number..

Algebraic

expression:
 $n + n = 2 \times n$

c

$$\frac{1}{100} = 1 \div 100$$

$$\frac{3}{20} = 3 \div 20$$

$$\frac{7}{3} = 7 \div 3$$

$$\frac{11}{7} = 11 \div 7$$

English

expression:

Any fraction is the numerator divided by the denominator

Algebraic

expression:
 $\frac{a}{b} = a \div b$

d

$$2 + (-5) = 2 - 5$$

$$4 + (-1) = 4 - 1$$

$$9 + (-12) = 9 - 12$$

$$(-3) + (-5) = (-3) - 5$$

English

expression:

Adding a negative number is the same as subtracting the positive.

Algebraic

expression:
 $a + (-b) = a - b$

e

$$2 \div 2 = 1$$

$$8 \div 8 = 1$$

$$6 \div 6 = 1$$

$$14 \div 14 = 1$$

English

expression:

Any number divided by itself is 1

Algebraic

expression:
 $n \div n = 1$

f

$$8 \div 1 = 8$$

$$5 \div 1 = 5$$

$$12 \div 1 = 12$$

$$13 \div 1 = 13$$

English

expression:

Any number divided by 1 is itself.

Algebraic

expression:
 $n \div 1 = n$

2

a

$$\begin{aligned}6 - 0 &= 6 \\1 - 0 &= 1 \\12 - 0 &= 12 \\17 - 0 &= 17\end{aligned}$$

Algebraic
expression:

$$n - 0 = n$$

b

$$\begin{aligned}3 + 4 &= 4 + 3 \\8 + 6 &= 6 + 8 \\16 + 7 &= 7 + 16 \\(-2) + 8 &= 8 + (-2)\end{aligned}$$

Algebraic
expression:

$$a + b = b + a$$

c

$$\begin{aligned}2 \times 3 &= 3 \times 2 \\4 \times 7 &= 7 \times 4 \\111 \times 56 &= 56 \times 111 \\(-2) \times 4 &= 4 \times (-2)\end{aligned}$$

Algebraic
expression:

$$a \times b = b \times a$$

d

$$\begin{aligned}2 + (-2) &= 0 \\3 + (-3) &= 0 \\15 + (-15) &= 0 \\8 + (-8) &= 0\end{aligned}$$

Algebraic
expression:

$$n + (-n) = 0$$

e

$$\begin{aligned}2 \times 2 &= 2^2 \\10 \times 10 &= 10^2 \\(-3) \times (-3) &= (-3)^2 \\24 \times 24 &= 24^2\end{aligned}$$

Algebraic
expression:

$$n \times n = n^2$$

f

$$\begin{aligned}7 \div 2 &= 7 \times \frac{1}{2} \\8 \div 4 &= 8 \times \frac{1}{4} \\11 \div \frac{2}{3} &= 11 \times \frac{3}{2} \\9 \div \frac{5}{4} &= 9 \times \frac{4}{5}\end{aligned}$$

Algebraic
expression:

$$a \div b = a \times \frac{1}{b}$$

3 Change these English expressions to algebraic expressions.

a Any number add zero equals itself.

$$n + 0 = n$$

b Any number added to itself is the same as multiplying that number by 2.

$$n + n = 2n$$

c Doubling any number twice is the same as multiplying it by 4.

$$n \times 2 \times 2 = 4n$$

d Multiplying any number by 5 is the same as multiplying it by 10, then dividing by 2.

$$n \times 5 = n \times 10 \div 2$$

Algebraic Conventions

② Apply algebraic conventions to simplify expressions

Leave out $\times$ symbol when multiplying.

$$3 \times x = 3x$$

Write numbers first, then variables.

$$x \times 6 = 6x$$

Hide the 1 when multiplying by 1 or -1 .

$$1 \times x = x$$

$$-1 \times x = -x$$

a $6 \times a$

b $t \times 3$

c $h \times z$

d $x \times 5 \times y$

$$6a$$

$$3t$$

$$hz$$

$$5xy$$

e $12 \times y \times x$

f $-1 \times y$

g $z \times 1$

h $h \times -1 \times n$

$$12yx$$

$$-y$$

$$z$$

$$-hn$$

Brackets without an operation in front means multiply. $3 \times 5 = 3(5)$

a 6×8

b 12×4

c -2×12

d -3×7

$$6(8)$$

$$12(4)$$

$$-2(12)$$

$$-3(7)$$

e $7 \times 4 \times 5$

f $12 \times (x - 3)$

g $(x - 5) \times -3$

h $-1 \times (x + 4)$

$$7(4)(5)$$

$$12(x - 3)$$

$$-3(x - 5)$$

$$-(x + 4)$$

Indices (powers) show repeated multiplication.

$$x \times x \times x \rightarrow x^3$$

a $w \times w$

b $9 \times w \times w$

c $z \times z \times z \times z$

d $x \times x \times w$

$$w^2$$

$$9w^2$$

$$z^4$$

$$x^2w$$

e $y \times 6 \times y$

f $y \times y \times z$

g $z \times y \times y$

h $x \times 7 \times x \times y$

$$6y^2$$

$$y^2z$$

$$zy^2$$

$$7x^2y$$

Show division using fractions.

$$x \div 3 = \frac{x}{3}$$

a $a \div 7$

b $7 \div a$

c $2x \div 13$

d $n \div y$

$$\frac{a}{7}$$

$$\frac{7}{a}$$

$$\frac{2x}{13}$$

$$\frac{n}{y}$$

Algebraic Conventions

Hide the $\times$ symbol when multiplying. $3 \times x = 3x$

Write numbers before pronumerals. $x \times 6 = 6x$

Hide 1 when multiplying by 1 or -1 . $1 \times x = x$

$$-1 \times x = -x$$

Brackets without an operation means multiply. $3 \times 5 = 3(5)$

Indices show repeated multiplication. $x \times x \times x = x^3$

Show division as fractions. $x \div 3 = \frac{x}{3}$

Key Ideas

1. Rules of reading and writing algebra are called algebraic
2. Using algebraic conventions make mathematical expressions,
so communication is more efficient.

1 Simplify these expressions by writing them without multiplication signs.

a $4 \times m$ $4m$	b $6 \times t$ $6t$	c $3 \times k$ $3k$	d $10 \times z$ $10z$
e $p \times 7$ $7p$	f $m \times 5$ $5m$	g $k \times 8$ $8k$	h $n \times 2$ $2n$
i $w \times z$ wz	j $c \times v$ cv	k $p \times q$ pq	l $a \times b$ ab
m $t \times t$ t^2	n $q \times q$ q^2	o $-1 \times s$ $-s$	p $a \times b \times c$ abc
q $2 \times s \times t$ $2st$	r $6 \times q \times p$ $6qp$	s $8 \times p \times q \times r$ $8pqr$	t $a \times a \times c$ a^2c
u $2 \times v \times v$ $2v^2$	v $5 \times w \times w$ $5w^2$	w $p \times p \times 8$ $8p^2$	x $p \times 8 \times p$ $8p^2$

2 Show the meaning of each algebraic expression by inserting multiplication signs.

a $9w$ $9 \times w$	b $5g$ $5 \times g$	c $52m$ $52 \times m$
d tz $t \times z$	e v^2 $v \times v$	f $3pq$ $3 \times p \times q$
g $8ab$ $8 \times a \times b$	h b^2 $b \times b$	i $12mn$ $12 \times m \times n$
j krz $k \times r \times z$	k $3abc$ $3 \times a \times b \times c$	l $35hs$ $35 \times h \times s$
m $-y$ $-1 \times y$	n $-4xy$ $-4 \times x \times y$	o $-x^3$ $-1 \times x \times x \times x$

3 Simplify these expressions by writing them as fractions.

a $y \div 6$	b $8 \div k$	c $6 \div w$
$\frac{y}{6}$	$\frac{8}{k}$	$\frac{6}{w}$
d $2 \div s$	e $9 \div t$	f $ab \div 3$
$\frac{2}{s}$	$\frac{9}{t}$	$\frac{ab}{3}$
g $pq \div 7$	h $3t \div 4$	i $7y \div 3$
$\frac{pq}{7}$	$\frac{3t}{4}$	$\frac{7y}{3}$
j $4k \div 9$	k $7tv \div 2$	l $12xy \div z$
$\frac{4k}{9}$	$\frac{7tv}{2}$	$\frac{12xy}{z}$
m $(z + 3) \div 2$	n $(3 + x) \div -4$	o $-11 \div (x + 1)$
$\frac{z + 3}{2}$	$\frac{3 + x}{-4}$	$\frac{-11}{x + 1}$

4 Show the meaning of each expression by rewriting them with division signs.

a $\frac{3}{10}$	b $\frac{m}{6}$	c $\frac{8}{n}$
$3 \div 10$	$m \div 6$	$8 \div n$
d $\frac{h}{3}$	e $\frac{z}{2}$	f $\frac{2}{p}$
$h \div 3$	$z \div 2$	$2 \div p$
g $\frac{pq}{2}$	h $\frac{km}{5}$	i $\frac{4}{5t}$
$pq \div 2$	$km \div 5$	$4 \div 5t$

5 Simplify these expressions using algebraic conventions.

a $4 \times m$ $4m$	b $6 \times t$ $6t$	c $m \times 5$ $5m$
d $x \times w$ xw	e $5 \times w \times w$ $5w^2$	f $4q \times (n + 5)$ $4q(n + 5)$
g $a \times 3 \times b$ $3ab$	h $h \div 3$ $\frac{h}{3}$	i $3 \div h$ $\frac{3}{h}$
j $a \times b \div 3$ $\frac{ab}{3}$	k $7 \div (3 + y)$ $\frac{7}{3 + y}$	l $(a + 3) \div 4$ $\frac{a + 3}{4}$
m $a \times b \times a \times a$ a^3b	n $c \times c \div 4$ $\frac{c^2}{4}$	o $(a - 4) \div (b + 9)$ $\frac{a - 4}{b + 9}$

6 Write the operations to show the difference between these three expressions.

a $2a$ $2 \times a$	b $\frac{2}{a}$ $2 \div a$	c $\frac{a}{2}$ $a \div 2$
d a^2 $a \times a$	e $2ab$ $2 \times a \times b$	f $(ab)^2$ $ab \times ab$

7 Show the meaning of each expression by writing it out in full.

a $9w$ $9 \times w$	b $48z$ $48 \times z$	c tz $t \times z$
d $3abc$ $3 \times a \times b \times c$	e $2w - 9$ $2 \times w - 9$	f $4(k - 9m)$ $4 \times (k - 9 \times m)$
g d^2 $d \times d$	h $3x^4$ $3 \times x \times x \times x \times x$	i $3ac^2d$ $3 \times a \times c \times c \times d$
j $\frac{1}{9}$ $1 \div 9$	k $\frac{h}{7}$ $h \div 7$	l $\frac{2a}{p}$ $2 \times a \div p$

Parts of an Expression

i Terms

- A **term** is part of an expression.
- Terms are **separated by addition**.
- Terms can be pronumerals, numbers, or a combination.

? Identify terms of an expression

$$3a + b + 13c$$

The terms are: $3a, b, 13c$

$$3xy + y - 12$$

$$= 3xy + y + (-12)$$

The terms are: $3xy, y, (-12)$

Identify the terms of these expressions

Expression	Terms	Expression	Terms
$3x + 9$		$9 - 3x - 4y$	
$3x - 9$		$9 - 3a - 4b$	
$3x + 4y - 9$		$3a^2$	
$3x - 4y - 9$		$3a^2 - 4$	
$3x - 4y + 9$		$3a^2 - 4b + 9$	
$-3x - 4y + 9$		$3a^2 - 4a + 9$	

💡 Key Ideas

1. Terms are parts of an algebraic expression that are separated by
2. In algebra, treat subtraction as adding the
3. In the expression $3x - 7$, the terms are: and

① Coefficients

- A **coefficient** is the number multiplying a pronumeral.

② Identify the coefficient of x

$3x$	$\frac{x}{3}$	$3 - x$	
Coefficient is 3	Coefficient is $\frac{1}{3}$	Coefficient is -1	
Identify the coefficient of x .			
a $3x$	b $5x$	c $13x$	d $2481x$
e $0.5x$	f $-123x$	g x	h $-x$
i $2x + y$	j $y + 2x$	k $y - 2x$	l $18 - 9x$
m y	n $2y$	o $\frac{1}{2}x$	p $\frac{x}{2}$
q $\frac{x}{5}$	r $\frac{3}{7}x$	s $\frac{3x}{7}$	t $x^2 + 5x$

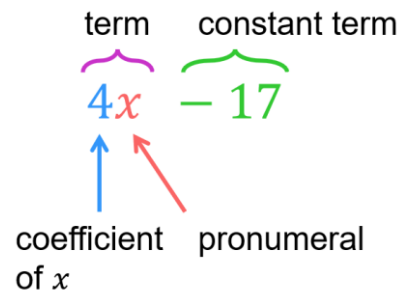
① Constant Term

- Constant terms** are numbers without variables.

② Identify the constant term

$3x + 5$	$-9x + 11$	$3x - 11$	
Constant is 5	Constant is 11	Constant is -11	
Identify the constant term			
a $3x + 6$	b $4x + 9$	c $12x + 7$	d $7 + 12x$
e $4x + 0.5$	f $4x + \frac{1}{2}$	g $4x - 1$	h $4x - 48$
i $y + 3x + 12$	j $2y - 3x + 4$	k $2xy - 9$	l $9y$

i Parts of an Expression



Example Identify the parts of an algebraic expression

$$2x - 4y - 9$$

Terms: $2x, -4y, -9$

Pronumerals: x, y

Coefficient of x : 2

Coefficient of y : -4

Constant term: -9

Guided Practice

a $3x - 6y + 14$

Terms:

$$3x, -6y, 14$$

Pronumerals:

$$x, y$$

Coefficient of x :

$$3$$

Coefficient of y :

$$-6$$

Constant term:

$$14$$

b $-2a + 4b + 9$

Terms:

$$-2a, 4b, 9$$

Pronumerals:

$$a, b$$

Coefficient of a :

$$-2$$

Coefficient of b :

$$4$$

Constant term:

$$9$$

c $17u + \frac{b}{2} - 12$

Terms:

$$17u, \frac{b}{2}, -12$$

Pronumerals:

$$u, b$$

Coefficient of u :

$$17$$

Coefficient of b :

$$\frac{1}{2}$$

Constant term:

$$-12$$

Key Ideas

1. A constant term is a term that has no
2. The coefficient is the number that a pronomeral is by.

1 List all the terms in these expressions.

a $3x + 2y$

$3x, 2y$

b $4a + 2b + c$

$4a, 2b, c$

c $5a + 3b + 2$

$5a, 3b, 2$

d $2x + 5$

$2x, 5$

2 Identify the constant terms in these expressions.

a $7a + 5$

5

b $2x + 3y + 6$

6

c $8 + 3x + 2y$

8

d $5a + 1 + 3b$

1

3 For each of the following expressions, state the coefficient of n .

a $17n + 24$

17

b $31 - 27a + 15n$

15

c $15w + 21n + 1$

21

d $15a + 2n + xy$

2

e $n + 51$

1

f $21 - n$

-1

g $0.5 - 8n$

-8

h $5 - \frac{3n}{4}$

$-\frac{3}{4}$

i

4 The expression $4x + 3y + 24z + 7$ has four terms.

a List all the terms.

$4x, 3y, 24z, 7$

b Identify the constant term.

7

c Identify the coefficient of x .

4

d Which pronumeral has coefficient 24?

z

5 Consider the expression $5a + 3b - 7c + 12$

a List all the terms.

$5a, 3b, -7c, 12$

b Identify the constant term.

12

c Identify the coefficient of c .

-7

6 Consider expression $3a + b + 13c$

a List all the terms.

$3a, b, 13c$

b Identify the constant term.

0

c Identify the coefficient of b .

1

7

Expression	Pronumerals	Coefficient of a	Constant Term
$3a + 9$	a	3	9
$3a - 9$	a	3	-9
$3b - 9$	b	3	-9
$3b$	b	3	0
$3a$	a	3	0
-9	None	0	-9
$2a + 3b - 9$	a, b	2	-9
$2a - 3b - 9$	a, b	2	-9
$2a - 3b + 9$	a, b	2	9
$-2a - 3b + 9$	a, b	-2	9
$9 - 2a - 3b$	a, b	-2	9
$a - 3b + 2$	a, b	1	2
$\frac{a}{3} - 3b + 2$	a, b	$\frac{1}{3}$	2
$\frac{2a}{3} - 3b + 2$	a, b	$\frac{2}{3}$	2
$-\frac{4a}{5} + 3b + 15$	a, b	$-\frac{4}{5}$	15
$3b - \frac{3a}{7} - 14$	a, b	$-\frac{3}{7}$	-14
$3a^2 + 4a + 1$	a	4	1
$3a^2 - 4a$	a	-4	0

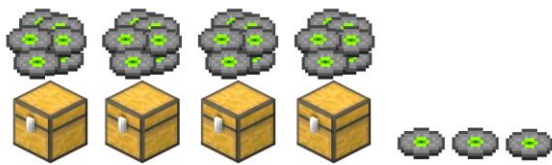
Forming Expressions from Diagrams

i Algebraic Expressions

- **Algebraic expressions** communicate relationships involving numbers.
- The **pronumeral** represents an unknown amount.
- The same **pronumeral** always represents the same number in one expression.

Example Represent a concrete scenario as an algebraic expression

Write an algebraic expression for the total number of music discs.



$$4n + 3$$

What does the pronumeral n represent?

n represents the of music discs in each chest.



$$2x + y + 1$$

Why did we need two pronumerals, x and y in this scenario?

One of the chests contains a number of music discs, so we must use a different pronumeral.

Guided Practice

Represent these scenarios as algebraic expressions



$$5x + 2$$



$$3x + y + 5$$



$$c + 2t$$



$$16 + 4r$$

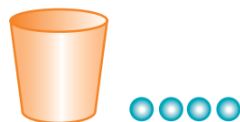


$$6 + 3a + b$$



$$40 + 2k + j$$

- 1 The diagram shows 1 cup and 4 marbles.



- a If the cup has 6 marbles, total marbles = $6 + 4 = \dots\dots$

$$6 + 4 = 10$$

- b If the cup has 20 marbles, total marbles = $\dots\dots + \dots\dots = \dots\dots$

$$20 + 4 = 24$$

- c If the cup has n marbles, total marbles = $\dots\dots\dots$

$$n + 4$$

- 2 The diagram shows 4 cups.



- a If each cup has 5 marbles, total marbles = $4 \times \dots = \dots\dots$

$$4 \times 5 = 20$$

- b If each cup has 8 marbles, total marbles = $4 \times \dots = \dots\dots$

$$4 \times 8 = 32$$

- c If each cup has n marbles, total marbles = $4 \times \dots = \dots\dots$

$$4 \times n = 4n$$

- 3 The diagram shows 3 cups and 2 marbles.



- a If each cup has 5 marbles, total marbles = $3 \times \dots + 2 = \dots\dots\dots$

$$3 \times 5 + 2 = 17$$

- b If each cup has 10 marbles, total marbles = $3 \times \dots + 2 = \dots\dots\dots$

$$3 \times 10 + 2 = 32$$

- c If each cup has n marbles, total marbles = $3 \times \dots + 2 = \dots\dots\dots$

$$3 \times n + 2 = 3n + 2$$

- 4 These two cups have a different number of marbles in them.

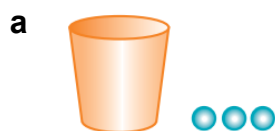
Explain why the total number of marbles is NOT $2n$



$2n$ means the two cups have the same number of marbles.

- 5 There are n marbles in each cup.

Write an algebraic expression for the total number of marbles in each diagram.



Total marbles = + 3

$n + 3$



Total marbles = $4 \times \dots\dots\dots = \dots\dots\dots$

$4 \times n = 4n$



$3n + 2$



$4n + 3$



$2n + 4$



$5n + 1$

- 6 There are x biscuits in each packet. Write an expression for the total number of biscuits.



$2x + 3$

- 7 There are x blocks in each bag. Write an expression for the total number of blocks.



$5x + 4$

Issy says:



the total number of blocks is $5x + 4$,
which is the same as $9x$.

Explain why she is incorrect.

There are not 9 bags of blocks.

- 8 x represents the number of marbles in a cup. Draw a diagram to represent:

a $4x + 2$

b $7x + 5$

- 9 Each cup contains x pills. Each bottle contains y pills.
Write the total number of pills in each scenario.



$x + y$



$x + 2y$



$x + 2y + 2$



$3y$



$3 + x + 2y$



$2x + 3y + 1$

g Draw $x + y + 5$

h Draw $5x + 2y + 2$

i Draw $x + 3y$

j Draw $3x + 2y + 4$

- 10 What expressions are represented by the question marks? Match the answers.

a

?		
x	x	5

b

x	x
?	5

c

5		
?	x	x

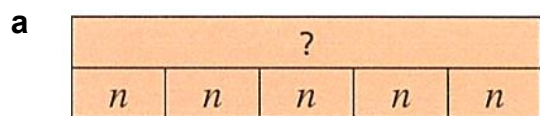
$2x - 5$

$5 - 2x$

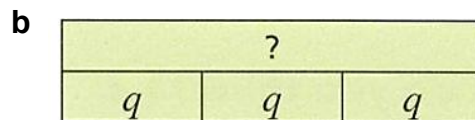
$2x + 5$

$a: 2x + 5, b: 2x - 5, c: 5 - 2x$

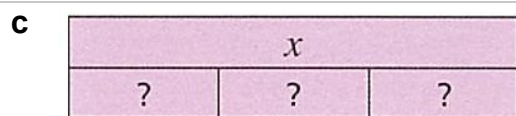
11 What expression is shown by the question mark in these bar models?



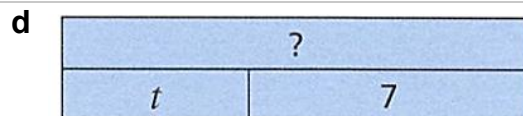
$5n$



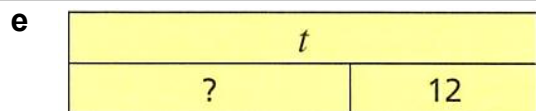
$3q$



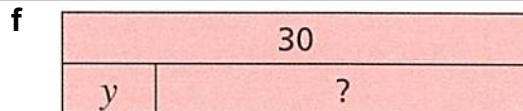
$3x$



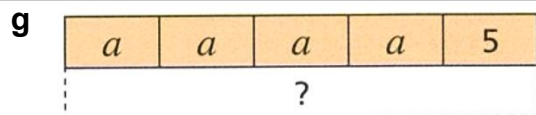
$t + 7$



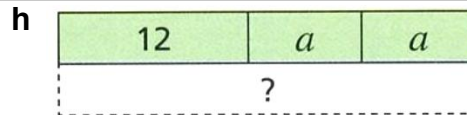
$t - 12$



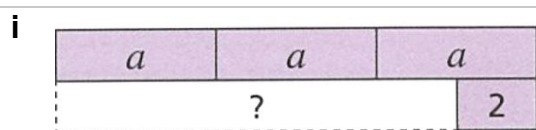
$30 - y$



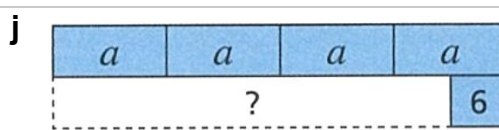
$4a + 5$



$2a + 12$



$3a - 2$



$4a - 6$

12 Draw bar models to represent these expressions.

a $b + 4$

b $b - 4$

c $\frac{b}{4}$

d $4b$

Forming Expressions from Descriptions

i The Language of Operations

Addition: combining two or more amounts.

- Sum
- Total
- More than
- Increased by
- Combined



Subtraction: removing an amount.

- Difference between
- Less than
- Decreased by



Multiplication: combining equal groups.

- Each
- Per
- Product
- Lots of
- Double ($\times 2$),
- Triple ($\times 3$),
- Quadruple...



Division: Splitting into equal groups.

- Each
- Per
- Divided by
- Shared between
- Quotient
- Out of
- Halve ($\div 2$),
- Third ($\div 3$),
- Quarter...



We often swap number order when we see "**than**" or "**from**"

"9 subtracted **from** 7" $7 - 9$

"12 less **than** 8" $8 - 12$

? Write algebraic expressions from a description

Sum of 5 and a

$$5 + a$$

Product of a and b

$$ab$$

3 less than a

$$a - 3$$

Represent these as an algebraic expression.

a 7 less than a

b a divided by 5

c b less than a

d 14 shared between n

e Product of x and 2

f n subtracted from 11

g Double n , then add 7

h Product of x and 2,
then divided by y

i n subtracted from 11,
then multiplied by 3

1 Write an expression for each of the following.

a The sum of 3 and 5 is+ the sum of p and q is + $3 + 5$ $p + q$	b 5 groups of 3 is $\times$ 5 groups of x is $\times$ 5×3 $5 \times x$
c The product of 2 and 6 is$\times$ the product of k and m is$\times$ 2×6 $k \times m$	d 5 more than 4 is+ 5 more than w is+ $5 + 4$ $5 + w$
e 2 less than 8 is- 2 less than c is- $8 - 2$ $c - 2$	f 9 squared is k squared is 9^2 k^2
g Double 7 is Double a is 2×7 $2a$	h Half of 10 is Half of b is $\frac{10}{2}$ $\frac{b}{2}$
i A third of 6 is A third of g is $\frac{6}{3}$ $\frac{g}{3}$	j A quarter of 12 is A quarter of h is $\frac{12}{4}$ $\frac{h}{4}$

2 Write an algebraic expression for each of the following.

a The sum of 2 and x $2 + x$	b The sum of ab and y $ab + y$	c 5 less than x $x - 5$
d The product of x and 3 $3x$	e One quarter of n $\frac{n}{4}$	f Half of s $\frac{s}{2}$
g The product of n and 9 $9n$	h y shared between 8 $\frac{y}{8}$	i t subtracted from 10 $10 - t$

3 Write an algebraic expression for each of the following.

d 5 is added to x , then the result is doubled. $2(5 + x)$	e a is tripled, then 4 is added. $3a + 4$	f 3 more than half of n $\frac{n}{2} + 3$
g 6 more than double n $2n + 6$	h The square of the sum of m and n $(m + n)^2$	i The sum of x and y is multiplied by 6. $6(x + y)$
j k is multiplied by 8, then 3 is subtracted. $8k - 3$	k The cube of the sum of m and n $(m + n)^3$	l The sum of x and y is divided by 5 $\frac{x + y}{5}$
m x is multiplied by 7 and the result is halved. $\frac{7x}{2}$	n k is multiplied by 8, then 3 is subtracted. $8k - 3$	o 10 less than the product of 4 and x $4x - 10$
p The square root of the sum of x and y $\sqrt{x + y}$	q The product of x and y is subtracted from 12. $12 - xy$	r 3 is subtracted from k , then the result is multiplied by 8. $8(k - 3)$

4 Write an expression for each of the following.

a 7 more than x $x + 7$	k x more than y $y + x$
b 7 less than x $x - 7$	l x multiplied by y xy
c 9 less than x $x - 9$	m x divided by y $\frac{x}{y}$
d 9 lots of x $9x$	n x divided by 3 $\frac{x}{3}$
e 19 lots of x $19x$	o x divided by 3 and then add 2 $\frac{x}{3} + 2$
f x divided by 19 $\frac{x}{19}$	p x divided by 3 and then subtract 2 $\frac{x}{3} - 2$
g x shared between 19 $\frac{x}{19}$	q x lots of 3 and then subtract 2 $3x - 2$
h x less than 19 $19 - x$	r x lots of 3 and then squared $(3x)^2$
i x less than 3 $3 - x$	s x lots of 3 squared 3^2x
j x less than y $y - x$	t x squared and then multiply by 3 $3x^2$

- 5 Write the calculation in numbers, then write the expression in algebra.

Using Numbers	Using Algebra
Becky begins a game with 12 marbles. She wins 3 and loses 5. How many marbles does she have at the end of the game? $12 + 3 - 5$	Becky begins a game with x marbles. She wins y and loses z . How many marbles does she have at the end of the game? $x + y - z$
In a class of 30 children, 10 have a sister. How many children don't have a sister? $30 - 10$	In a class of a children, b have a sister. How many children don't have a sister? $a - b$
There were 30 questions in a test. Megan got 25 correct. How many did she get wrong? $30 - 25$	There were x questions in a test. Megan got y correct. How many did she get wrong? $x - y$
Lucy went on a journey, going 25 kilometres by train, 4 kilometres by bus, and 2 kilometres on foot. What is the total distance she travelled? $25 + 4 + 2$	Lucy went on a journey, going x kilometres by train, y kilometres by bus, and z kilometres on foot. What is the total distance she travelled? $x + y + z$
Mark buys 7 chocolate bars, each costing 3 dollars. How much does he spend in total? 7×3	Mark buys a chocolate bars, each costing b dollars. How much does he spend in total? ab
Maddie has 5 cards. Emma has 2 times as many as Maddie. How many cards does Emma have? 5×2	Maddie has a cards. Emma has b times as many as Maddie. How many cards does Emma have? ab

6 Write an expression for each of the following:	
a There are 12 people on a bus. x get off. How many people remain?	b A train has 88 passengers. y people get on. How many are on the train?
$12 - x$	$88 + y$
c A can of lemonade costs c dollars. How much do 5 cans of lemonade cost?	d Each ice block is \$3. Each lolly bag is \$2 How much money is earned from selling x ice blocks and y lolly bags?
$5c$	$3x + 2y$
e A rope is x metres long. I cut off four lengths of y metres each. How much rope remains?	f n students go on an excursion. How much should Ms Ali charge each student if she wants to share the bus hire fee of \$200?
$x - 4y$	$\frac{200}{n}$
7 Nicholas buys 10 lolly bags from a supermarket.	
a If there are 7 lollies in each bag, how many lollies does he buy in total?	b If there are n lollies in each bag, how many lollies does he buy in total?
$7 \times 10 = 70$	$10n$
8 Write an algebraic expression for the cost of:	
a x pies at \$6 each.	b 6 pies at \$ x each.
$6x$	$6x$
c 10 pies at \$ n each.	d y drinks at \$2 each.
$10n$	$2y$
e x pies at \$5 and y drinks at \$2	f x pies at \$2 and y drinks at \$6
$5x + 2y$	$2x + 6y$
9 The cost of 10 cans of drink is \$ x . Write an expression for:	
a The cost of one can of drink	b The cost of n cans
$\frac{x}{10}$	$\frac{nx}{10}$

10 If pencils cost $\$x$ each, write an expression for the cost of:

a 10 pencils

$$10x$$

b k pencils

$$kx$$

c 3 packets of pencils,
if each packet contains 5
pencils

$$15x$$

11 Recall that there are 100 centimetres in 1 metre and 1000 metres in 1 kilometre.

Write expressions for each of the following.

a How many metres are
there in x km?

$$1000x$$

b How many centimetres
are there in x metres?

$$100x$$

c How many centimetres
are there in x km?

$$100\,000x$$

12

a Ed is 12 years old. How old will he be in 2 years' time?

$$12 + 2 = 14$$

b Faith is y years old. How old will she be:

i in 2 years' time

$$y + 2$$

ii in x years' time?

$$y + x$$

iii 5 years ago?

$$y - 5$$

12 A rectangular garden bed is 12 m long and 5 m wide.

a Find the area of the garden bed. Show your working out.

$$60 \text{ m}^2$$

b The length is increased by x cm and the width are decreased by y cm.
Find the new length and width of the garden.

$$\begin{array}{l} \text{length: } 12 + x \\ \text{width: } 5 - y \end{array}$$

c Write an expression for the area of the new garden bed.

$$(12 + x)(5 - y)$$

13 Mikayla is paid $\$x$ per hour at her job. Write an expression for each of the following:

a How much does Mikayla earn if she works 8 hours?

$$8x$$

b Mikayla gets a pay rise of $\$3$ per hour, what is her new hourly wage?

$$x + 3$$

c Mikayla works 8 hours at the increased hourly rate, write how much she earns as an expression.

$$8(x + 3)$$

14 Match up each sentence to its algebraic expression.

Multiply b by 2 then subtract from a	a	A $a(b - c)$
Subtract c from a then multiply by b	b	B $4a - c$
Multiply a by b then divide by c	c	C $\frac{c}{a + b}$
Add b to a then divide into c	d	D $\frac{ab}{c}$
Divide c by b then multiply by a	e	E abc
Multiply a by c then divide into b	f	F $b(a - c)$
Add b to a then divide by c	g	G $a - 2b$
Multiply ab by c	h	H $\frac{b}{ac}$
Multiply a by 4 then subtract c	i	I $\frac{a + b}{c}$
Subtract c from b then multiply by a	j	J $\frac{ac}{b}$

$a - G,$
 $b - F,$
 $c - D,$
 $d - C,$
 $e - J,$
 $f - H,$
 $g - I,$
 $h - E,$
 $i - B,$
 $j - A$

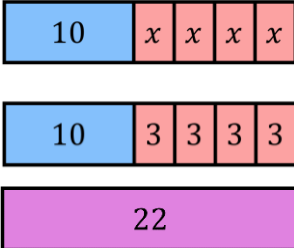
Substitution and Evaluation

① Substitution and Evaluation

- To **substitute** means to **replace** pronumerals with numbers.
- To **evaluate** means to work out the final number by following the **order of operations**.

🔧 Example Substitute and evaluate expressions

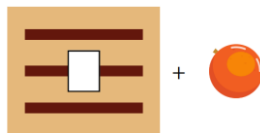
The bar model shows $10 + 4x$. Evaluate the expression when $x = 3$.

$$\begin{aligned}
 &10 + 4x \\
 &= 10 + 4(3) \\
 &= 10 + 12 \\
 &= 22
 \end{aligned}$$


✍ Guided Practice

- a** The total number of oranges is $x + 1$
Evaluate the expression for $x = 5$

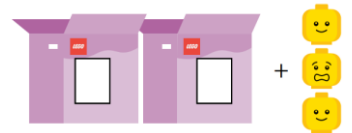
$$\begin{aligned}
 &x + 1 \\
 &= \dots + 1 \\
 &= \dots
 \end{aligned}$$



6

- b** The number of Lego pieces is $2x + 3$
Evaluate the expression for $x = 6$

$$\begin{aligned}
 &2x + 3 \\
 &= \dots \\
 &= \dots
 \end{aligned}$$



15

- c** Evaluate $7 + 5b$ when $b = 5$

32

- d** Evaluate $\frac{x}{2} + 5$ when $x = 12$

11

- e** Evaluate $\frac{n+7}{3}$ when $n = 8$

5

- f** Evaluate $7 - 4x$ when $x = 2$

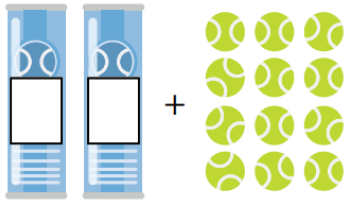
-1

💡 Key Ideas

- Substitution means to a letter with a number.
- means to calculate the numerical value of an expression.
- When evaluating, always follow the

1

a The diagram shows $2t + 12$ tennis balls.



Evaluate when $t = 6$.

$$\begin{aligned} &2t + 12 \\ &= 2(\quad) + 12 \\ &= \dots\dots\dots \end{aligned}$$

24

b The diagram shows $3c + 4$ coins.

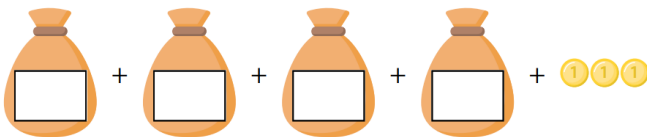


Evaluate when $c = 10$.

$$\begin{aligned} &3c + 4 \\ &= 3(\quad) + 4 \\ &= \dots\dots\dots \end{aligned}$$

34

c The diagram shows $4x + 3$ coins.



How many coins are there if $x = 12$?

51

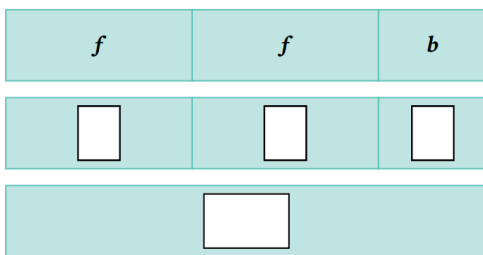
d The diagram shows $2c + 3$ cars.



How many cars are there if $c = 6$?

15

2 The bar model shows $2f + b$. Evaluate when $f = 7$, $b = 3$



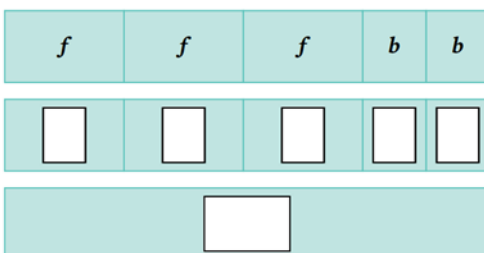
$$2f + b$$

$$= 2(\quad) + (\quad)$$

$$= \dots\dots\dots$$

17

3 The bar model shows $3f + 2b$. Evaluate when $f = 7$, $b = 3$



$$3f + 2b$$

$$= 3(\quad) + 2(\quad)$$

$$= \dots\dots\dots$$

27

4 Substitute $c = 5$ to evaluate each expression.

a $c + 10$	b $10 + c$	c $10c$	d $\frac{c}{10}$	e $\frac{10}{c}$
15	15	50	0.5	2

5 Substitute $x = 5$ into these expressions and evaluate the result. Show the substitution step.

a $x + 3$ $= (5) + 3$ $= \dots\dots\dots$	b $14 - x$	c $2x$
8	9	10
d $2x + 4$	e $13 - 2x$	f $3x - 6$
14	3	9
d $10x$	e $10 + x$	f $\frac{x}{10}$
50	15	$\frac{1}{2}$
d $x - 10$	e $10 - x$	f x^2
-5	5	25

6 Evaluate these expressions for $g = 400$.

a $\frac{g}{10}$	b $\frac{10}{g}$	c $\frac{g}{100}$	d $\frac{100}{g}$	e $\frac{g}{20}$	f $\frac{20}{g}$
40	$\frac{1}{40}$	4	$\frac{1}{4}$	20	$\frac{1}{20}$
g $\frac{g}{200}$	h $\frac{200}{g}$	i $\frac{g}{400}$	j $\frac{400}{g}$	k $\frac{g}{1000}$	l $\frac{1000}{g}$
2	$\frac{1}{2}$	1	1	$\frac{2}{5}$	$\frac{5}{2}$

7 Work out the value of each of these expressions when $a = 5$

a $10a + 3$	b $\frac{10}{a} + 3$	c $10a - 3$	d $\frac{a}{10} + 3$
53	5	47	3.5
e $10(a + 3)$	f $\frac{10}{a + 3}$	g $10(a - 3)$	h $3 + \frac{a}{10}$
80	$\frac{5}{4}$	20	3.5
i $\frac{10 - a}{3}$	j $10 - \frac{3}{a}$	k $3 + \frac{10}{a}$	l $3 - \frac{10}{a}$
$\frac{5}{3}$	$\frac{47}{5}$	5	1

8 If $c = 3$, evaluate these expressions.

a $4 + c$	h $\frac{5c - 7}{2}$
	7
b $4c$	i $\frac{4c}{4}$
	12
c $4c + c$	j $\frac{c}{4c}$
	15
d $5c$	k c^2
	15
e $5c - 7$	l $2c^2$
	8
f $(5c - 7) \div 2$	m $(2c)^2$
	4
g $5c - 7 \div 2$	n $\frac{4c}{4}$
	11.5
	4
	3

9 Evaluate these expressions when $g = 80$.

a $\frac{g}{10} + 5$	b $\frac{g}{10} - 5$	c $10g + 5$	d $5 + \frac{g}{10}$	e $10g - 5$
13	3	805	13	795
f $10 - \frac{g}{20}$	g $10 - \frac{20}{g}$	h $10 + \frac{40}{g}$	i $\frac{10}{g} + 40$	j $g^2 - 1400$
6	$9\frac{3}{4}$	10.5	$40\frac{1}{8}$	5000

10 Find the values of each of these expressions when $c = 100$

a $\frac{c}{5} + 1$	b $\frac{c+1}{5}$	c $\frac{c}{4} + 1$	d $\frac{c+1}{4}$	e $\frac{c}{10} + 1$	f $\frac{c+1}{10}$
21	$\frac{101}{5}$	26	$\frac{101}{4}$	11	$\frac{101}{10}$
g $\frac{c+10}{5}$	h $\frac{c}{5} + 10$	i $\frac{c^2+10}{5}$	j $\frac{c^2}{5} + 10$	k $\frac{c^2-10}{5}$	l $\frac{c^2}{5} - 10$
22	30	2002	2010	1998	1990
m $\sqrt{c}$	n $10\sqrt{c}$	o $\frac{\sqrt{c}}{5}$	p $\sqrt{c} + 5$	q $5 + \sqrt{c}$	r $1 + 5\sqrt{c}$
10	100	2	15	15	51

11 Work out the value of each of these expressions when $x = 3$ and $y = 0$

a $x + y$	b $x + 2y$	c $x - y$	d $x - 2y$
3	3	3	3
e xy	f $\frac{y}{x}$	g $\frac{x}{2+y}$	h $\frac{x}{2-y}$
0	0	1.5	1.5
i $x^2 + y$	j $(x+y)^2$	k $x + y^2$	l $x(y+2)$
9	9	3	6

12 If $m = 9$ and $n = 16$, find the values of:

a $\sqrt{m} + \sqrt{n}$

7

b $\sqrt{m + n}$

5

c $m\sqrt{n}$

36

d $n\sqrt{m}$

48

e $\sqrt{mn}$

12

13 x and y are two numbers, where $xy = 24$.

a What values could x and y equal if they are whole numbers?

Try to list as many as possible.

b What values could x and y be if they can be decimals, fractions, or whole numbers?

14 Write two different expressions involving x that evaluate to -10 if $x = -5$.

15

a Find the difference between $4a + 3$ and $4(a + 3)$ when $a = 2$

b Find the difference between $4a + 3$ and $4(a + 3)$ when $a = 10$

c Find the difference between $4a + 3$ and $4(a + 3)$ when $a = 100$

d Find the difference between $4a + 3$ and $4(a + 3)$ when $a = 0.2$

e What do you notice?

f Without any calculations, find the difference between $4a + 3$ and $4(a + 3)$ when $a = 786.174$

16 Complete the table.

a	10	0	2	12	5	1
$a + 2$	12	2	4	14	7	3
$2a$	20	0	4	24	10	2
a^2	100	0	4	144	25	1
$2 - a$	-8	2	0	-10	-3	1
$\frac{a}{2}$	5	0	1	6	$\frac{5}{2}$	0.5

17 Complete the table.

x	5	10	7	9	5	8	6
y	3	4	2	5	2	0	-6
$x + y$	8	14	9	14	7	8	0
$x - y$	2	6	5	4	3	8	12
xy	15	40	14	45	10	0	-36

18 Evaluate each expression for $n = 10$.

$$n = 10$$

$$5n =$$

$$5n + 2 =$$

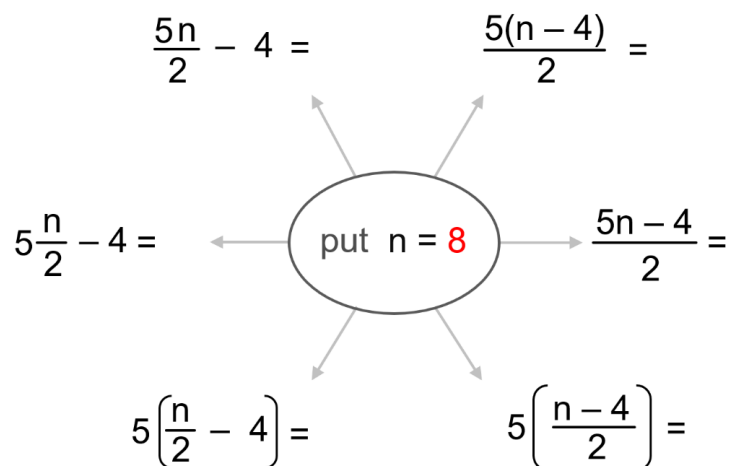
$$\frac{5n + 2}{4} =$$

$$3 \left(\frac{5n + 2}{4} \right) = \frac{3 \left(\frac{5n + 2}{4} \right) + 6}{5} = 7 \left(\frac{3 \left(\frac{5n + 2}{4} \right) + 6}{5} \right) - 1 =$$

$$3 \left(\frac{5n + 2}{4} \right) + 6 =$$

$$\frac{7 \left(\frac{3 \left(\frac{5n + 2}{4} \right) + 6}{5} \right) - 1}{2} =$$

- 19 Substitute $n = 8$ into each expression and evaluate the result.



Which of the expressions are equivalent?

$$\frac{5n}{2} - 4 \text{ and } 5\frac{n}{2} - 4,$$

$$\frac{5(n-4)}{2} \text{ and } 5\left(\frac{n-4}{2}\right)$$

20 **NAPLAN B**

Evaluate the expression $x^2 + 3x - 4$ when $x = 3$

14

- 21 $R = 2d - 3e - 4f$ Find the value of R if $d = 2, e = 3, f = \frac{1}{2}$

-7

- 22 Evaluate $\frac{20}{y} + 3x$ when $x = \frac{1}{5}, y = 2$

$10\frac{3}{5}$

- 23 Evaluate $\frac{20}{x} + 3y$ when $x = \frac{1}{5}, y = 2$

24 Evaluate the expressions for the given value of n

a

$$n = 5$$

$$3n =$$

$$3n + 10 =$$

$$2(3n + 10) =$$

$$2(3n + 10) - 40 =$$

$$6(2(3n + 10) - 40) =$$

$$6(2(3n + 10) - 40) - 55 =$$

5

b

$$n = 2$$

$$5n =$$

$$5n + 4 =$$

$$2(5n + 4) =$$

$$2(5n + 4) - 18 =$$

$$3(2(5n + 4) - 18) =$$

$$3(2(5n + 4) - 18) - 28 =$$

2

c

$$n = 5$$

$$2n - 4 =$$

$$\frac{2n - 4}{3} =$$

$$7 \left(\frac{2n - 4}{3} \right) =$$

14

d

$$n = 9$$

$$3n + 8 =$$

$$\frac{3n + 8}{5} =$$

$$2 \left(\frac{3n + 8}{5} \right) =$$

14

25 Find the value of n by working backwards.

$$n =$$

$$5n =$$

$$5n + 2 =$$

$$\frac{5n + 2}{4} =$$

$$3 \left(\frac{5n + 2}{4} \right) =$$

$$3 \left(\frac{5n + 2}{4} \right) + 6 =$$

$$7 \left(\frac{3 \left(\frac{5n + 2}{4} \right) + 6}{5} \right) =$$

$$\frac{3 \left(\frac{5n + 2}{4} \right) + 6}{5} =$$

$$7 \left(\frac{3 \left(\frac{5n + 2}{4} \right) + 6}{5} \right) - 1 =$$

$$\frac{7 \left(\frac{3 \left(\frac{5n + 2}{4} \right) + 6}{5} \right) - 1}{2} = 10$$

$n = 2$

Substitution with Negative Numbers

Substituting Negative Numbers

- **Always put brackets around negative numbers when substituting.**
- Without brackets, we might mistake a negative sign for subtraction.
- Brackets help calculators follow the order of operations, especially when raising negative numbers to a power.

Example Evaluate expressions after substituting negative integers

Evaluate:

$10 + 4b \text{ for } b = -2$

$10 + 4b$

$= 10 + 4(-2)$

$= 10 + (-8)$

$= 2$

$12 - 3b \text{ for } b = -5$

$12 - 3b$

$= 12 - 3(-5)$

$= 12 - (-15)$

$= 27$

$2b^2 \text{ for } b = -3$

$2b^2$

$= 2(-3)^2$

$= 2(9)$

$= 18$

Guided Practice

a Evaluate $x + 1$ for $x = -5$

$x + 1$

$= \dots + 1$

$= \dots$

-4

b Evaluate $2x + 3$ for $x = -3$

$2x + 3$

$= \dots$

$= \dots$

-3

c Evaluate $4b^2$ for $b = -3$

36

d Evaluate $4 - x$ for $x = -3$

7

e Evaluate $\frac{2x+7}{3}$ for $x = -4$

$-\frac{1}{3}$

f Evaluate $12 - 3b$ for $b = -5$

27

Key Ideas

1. Always substitute negative numbers in so you are less likely to make a mistake.

1 For $x = -5$, evaluate these expressions.

a $x + 3$ $= (-5) + 3$ $= \dots\dots\dots$	b $14 - x$	c $2x$
-2	19	-10
d $2x + 4$	e $13 - 2x$	f $3x - 6$
-6	23	-21
d $\frac{5x}{7}$	e $-3x + 15$	f $10x - 50$
$-\frac{25}{7}$	30	-100

2 For $a = 6$ and $b = -2$, evaluate these expressions.

a $a + b$	b $a - b$	c $b - a$
4	8	-8
d $2a + b$	e $5b + 2a$	f $6b - 7a$
10	2	-54

3 Flo is working out the value of $c - d$. She is told that $c = 4$ and $d = -9$. She writes:

$$\begin{aligned} c - d \\ = 4 - 9 \\ = -5 \end{aligned}$$

Explain why she is wrong.

She substituted 9, not -9

Provide the correct answer.

4 If $m = -4$ and $n = 20$, find the value of:

a $m + 9$

5

b $n - 25$

-5

c $7 + m$

3

d $m + n$

16

e $n - m$

24

f $m - n$

-24

g $2m$

-8

h $2n$

40

i $-2m$

8

j mn

-80

k n^2

400

l m^2

16

5 If $p = 4$, $q = -3$, $r = 5$, work out the value of:

a $pq + qr$

-27

b $pq - qr$

3

c $p^2 + q^2$

25

d $(r - q) \div (p - 6)$

-4

6 $f = -3$, $g = -10$, $h = 9$. Find the value of:

a $2f + h$

3

b $3f + h$

0

c $h - g$

19

d $f - 2g$

17

e fg

30

f gh

-90

g fgh

270

h $fg - 20$

10

7 For $m = 3$ and $n = 7$, find the value of:

a $\sqrt{6m + n}$

5

b $\sqrt{10(m + n)}$

10

c $\sqrt{4(n - m)}$

4

d $\sqrt{7m + 4n}$

7

8 For $x = 3$, $y = \frac{1}{2}$, $z = -5$ evaluate these expressions.

a	$x + 4$		k	$-2z$	10
		7	l	$-2z - x$	7
b	$x - 4$		m	$-2z - x - y$	6.5
		-1	n	$2(-2z - x - y)$	13
c	$4x$		o	$2(2z + x + y)$	-13
		12	p	$2x + x + y$	9.5
d	$4x + y$		q	$\frac{2x+x+y}{2}$	4.75
		12.5	r	$\frac{y+x+2z}{2}$	-3.25
e	$4x + 2y$		s	$\frac{y+x-2z}{2}$	6.75
		13	t	$y + x - \left(\frac{2z}{2}\right)$	8.5
f	$4x - 2y$				
		11			
g	$2y - 4x$				
		-11			
h	$2y + z$				
		-4			
i	$3x + z$				
		4			
j	$3x - z$				
		14			

9 Evaluate these expressions by substituting $a = -2$, $b = 6$, $c = -3$

a	$a^2 - b$		b	$a - b^2$		c	$2c + a$	
		-2			-38			-8
d	$b^2 - c^2$		e	$a^3 + c^2$		f	$3ab + ac$	
		27			1			-30
g	$c - 2ab$		h	$abc - (ac)^2$		i	$c(a + b)$	
		21			0			-12

10 $p = 8$ and $q = -12$. Work out

a $p + q$	b $q - p$	c pq
-4	-20	-96
d $p - 2q$	e $\frac{2q}{p}$	f $\frac{pq}{2}$
32	-3	-48
g $p - \frac{q}{2}$	h $\frac{p-q}{2}$	i $\frac{q-p}{4}$
14	10	-5

11 2008 HSC Standard 2 Band 4

What is the value of $\sqrt{\frac{x+2y}{8y}}$ if $x = 5.6$ and $y = 3.1$, correct to 2 decimal places?

- (A) 0.69
- (B) 2.62
- (C) 2.83
- (D) 4.77

A

12 HSC Standard 2 Sample Question Band 4

What is the value of $\frac{a+b}{ab}$ if $a = -2.1$ and $b = -3.6$, correct to 1 decimal place?

-0.8

13 2010 HSC Standard 2 Band 5

If $M = -9$, what is the value of $\frac{3M^2+5M}{6}$?

- (A) -250.5
- (B) -48
- (C) 33
- (D) 235.5

C

14 Evaluate $8p - \frac{14}{q}$ when $p = \frac{-3}{8}$, $q = -7$

-1

15 Complete the table.

x	3	1	0.25	-2	2	$\frac{8}{7}$
$4x + 2$	14	6	3	-6	10	$\frac{46}{7}$
$4 - 3x$	-5	1	3.25	10	-2	$\frac{4}{7}$
$2x - 4$	2	-2	-3.5	-8	0	$-\frac{12}{7}$

16 Complete the table.

a	5	8	2	3	-20	10
b	2	2	1	7	10	-3
$a + b$	7	10	3	10	-10	7
$a + 2b$	9	12	4	17	0	4
$a - b$	3	6	1	-4	-30	13

Generating Patterns by Substitution

① Consecutive Numbers Create Patterns

Substituting **consecutive** numbers into an expression can make a **number pattern**.

🔗 Example Generate a number pattern by substitution

x	0	1	2	3	4	5		10		100	Change by 1, starting at 3
$x + 3$	3	4	5	6	7	8		13		103	

x	0	1	2	3	4	5		10		100	Change by 2, starting at -1
$2x - 1$	-1	1	3	5	7	9		19		199	

✎ Guided Practice

Complete these tables to generate a number pattern from the algebraic expression.

a

x	0	1	2	3	4	5		10		100	Change by Starting at
$2x$											

0 2 4 6 8 10 20 200; change by 2 starting at 0

b

x	0	1	2	3	4	5		10		100	Change by Starting at
$2x + 3$											

3 5 7 9 11 13 23 203 change by 2 starting at 3

c

x	0	1	2	3	4	5		10		100	Change by Starting at
$5x - 7$											

-7 -2 3 8 13 18 43 493 change by 5 starting at -7

d

x	0	1	2	3	4	5		10		100	Change by Starting at
$-4x$											

0 -4 -8 -12 -16 -20 -40 -400 change by -4 starting at 0

💡 Key Ideas

1. By substituting numbers, we can form a number pattern.

1 Complete these tables to generate a number pattern from the algebraic expression.

a

x	0	1	2	3	4	5		10		100
$3x$										

Change by
Starting at

0 3 6 9 12 15 30 300 change by 3 starting at 0

b

x	0	1	2	3	4	5		10		100
$3x + 1$										

Change by
Starting at

1 4 7 10 13 16 31 301 change by 3 starting at 1

c

x	0	1	2	3	4	5		10		100
$3x + 2$										

Change by
Starting at

2 5 8 11 14 17 32 302 change by 3 starting at 2

d

x	0	1	2	3	4	5		10		100
$2 + 3x$										

Change by
Starting at

2 5 8 11 14 17 32 302 change by 3 starting at 2

e

x	0	1	2	3	4	5		10		100
$-3x + 2$										

Change by
Starting at

2 -1 -4 -7 -10 -13 -28 -298 change by -3 starting at 2

f

x	0	1	2	3	4	5		10		100
$2 - 3x$										

Change by
Starting at

2 -1 -4 -7 -10 -13 -28 -298 change by -3 starting at 2

g

x	0	1	2	3	4	5		10		100
$-2x + 2$										

Change by

Starting at

0 -2 -4 -6 -8 -18 -198 change by -2 start at 2

h

x	0	1	2	3	4	5		10		100
$-x - 2$										

Change by

Starting at

-2 -3 -4 -5 -6 -7 -12 -102 change by -1 starting at -2

i

x	0	1	2	3	4	5		10		100
$-x + 1$										

Change by

Starting at

1 0 -1 -2 -3 -4 -9 -99 change by -1 starting at 1

j

x	0	1	2	3	4	5		10		100
$1 - x$										

Change by

Starting at

1 0 -1 -2 -3 -4 -9 -99 change by -1 starting at 1

k

x	0	1	2	3	4	5		10		100
$-x$										

Change by

Starting at

0 -1 -2 -3 -4 -5 -10 -100 change by -1 starting at 0

l

x	0	1	2	3	4	5		10		100
$3x - 5$										

Change by

Starting at

-5 -2 1 4 7 10 25 295 change by 3 starting at -5

2 By looking at the questions in **Q1**, complete the sentences:

In the expression $mx + c$: the coefficient of x is and the constant term is

The coefficient of x represents how much to change as you increase x by

The constant term represents the number to at, i.e., when $x = 0$

3 Determine the number pattern formed by these equations.

a $3x + 6$ Change by 3 Starting at 6	b $3x + 9$ Change by Starting at Change by 3 starting at 9	c $6x + 18$ Change by Starting at Change by 6 starting at 18
d $3x + 18$ Change by Starting at Change by 3 starting at 18	e $3x - 18$ Change by Starting at Change by 3 starting at -18	f $2x - 18$ Change by Starting at Change by 2 starting at -18
g $18 - 2x$ Change by Starting at Change by -2 starting at 18	h $-2x + 18$ Change by Starting at Change by -2 starting at 18	i $-4x + 18$ Change by Starting at Change by -4 starting at 18
j $-4x - 18$ Change by Starting at Change by -4 starting at -18	k $4x - 18$ Change by Starting at Change by 4 starting at -18	l $-5.4 - \frac{9}{4}x$ Change by Starting at Change by $-\frac{9}{4}$ starting at -5.4